

Dhaneswara Mandrasa Triweko

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PROFESSIONAL SUMMARY

Decision scientist with 8+ years in data science across telecom, banking, agriculture, geospatial and commerce, building causal models and LLM-powered data products; graduate research fuses LLM-derived features with causal inference and discrete choice models. Built models at 100M-user scale, a telco credit score with the company's best-ever Gini (0.6), and recommendation systems that lifted cross-sell revenue 12%. Currently estimating how congestion pricing shifts ride-hailing demand, combining quasi-experimental identification with behavioural choice models. Looking for principal-track individual-contributor work where the modelling informs pricing, marketplace and policy decisions.

RESEARCH

- Master's thesis (in progress), Urban and Data Science Lab, Hiroshima University**2026 – present
- LLM feature pipeline:** Llama-3 70B scoring of OpenStreetMap POIs and place names into 15 structured semantic features across 4 construct families (activity, physical environment, safety, built environment).
 - Prompt engineering and reliability:** versioned prompt library with staged probes to select wordings; repeated-run reproducibility testing; scorer chosen by training cutoff to prevent leakage from the policy period.
 - Causal evaluation:** difference-in-differences on millions of NYC TLC high-volume for-hire vehicle (HVFHV) trip records around NYC's January 2025 congestion toll.
 - Ablation design:** 4-arm ladder (no-LLM baseline, flat LLM features, latent-factor ICLV, embeddings) to measure what the LLM features add, with a SHAP-explained gradient-boosting robustness check.
 - Hybrid choice modelling:** factor analysis (N=262 zones) reduces the features to latent factors for an ICLV model; welfare via logsum consumer surplus, with price elasticity.
 - Pipeline engineering:** medallion layers (bronze, silver, gold) and multi-agent LLM tooling to coordinate the workflow.

EDUCATION

- Hiroshima University**Hiroshima, Japan
*MPhil, Smart Mobility (AI and data science for transport)*Apr. 2026 – Mar. 2028 (expected)
Urban and Data Science Lab, supervised by Prof. Tao Feng. Programme: Innovation and Practice for Smart Society.
- Institut Teknologi Bandung (ITB)**Bandung, Indonesia
*B.Sc. Geological Engineering – GPA 3.52/4.00, Cum Laude*Aug. 2011 – Aug. 2015

PROFESSIONAL EXPERIENCE

- Olsera – cloud POS & commerce platform for MSMEs**Jakarta, Indonesia
Head of Data & AIAug. 2025 – Mar. 2026
 - Owned data & AI strategy with a team of 1 data scientist and 1 data engineer, reporting to executives.
 - Designed the data architecture and schemas (orders, products, roles, AI stock recommendations, logs) for analytics and AI use cases.
 - Ran POC evaluations of Snowflake vs Databricks, benchmarked against BigQuery and Redshift, to choose the company's data platform.
- iZeno – enterprise IT & data consulting**Jakarta, Indonesia
Lead Data ScientistJul. 2024 – Mar. 2025
 - Bank BRI:** designed a hybrid recommendation engine (collaborative filtering + XGBoost) that improved product-targeting accuracy by 35% and drove a 12% increase in cross-sell revenue.
 - Sinarmas:** led data scientists, agronomists and DevOps in a root-cause analysis system in Dataiku that increased palm oil yield by 18% and cut waste by \$1.2M/year.
 - Labuan:** architected an LLM-driven news analytics system on Snowflake Cortex, reducing misclassification by 27%, with a knowledge graph of 10K+ entities serving 50+ stakeholders.
 - Led a team of 2 data scientists; client engagements improved operational efficiency by 25–40%.
- Eureka AI – telecom data monetization, 7 countries**Abu Dhabi, UAE
Senior Data ScientistApr. 2021 – Jul. 2024
 - Built a telco- and clickstream-based credit score model reaching a Gini of 0.6, the highest in the company's history at the time.
 - Segmented 100M+ daily active ride-hailing and e-commerce users into hierarchical segments across five major Asian telecom operators, using clickstream data with k-means and ensemble models (Random Forest, XGBoost) in PySpark ML and MLflow.
 - Made model decisions explainable to business users with SHAP and other model-agnostic interpretation methods.

- Classified 30+ personas for 80M+ monthly active subscribers from clickstream, mobility and subscriber data; built a Word2Vec domain look-alike model in Spark.
- Led Market Intelligence products that grew from 5% to 70% of company revenue in under two years.

Bhumi Varta Technology – location intelligence

Jakarta, Indonesia

Data Scientist

Jun. 2020 – Apr. 2021

- Built spatial models on POI, mobile and sales data that guided clients' expansion from under 100 to 200+ outlets across Indonesia.
- Estimated poverty levels from satellite imagery with CNNs (PyTorch), refining socio-economic status resolution from 1×1 km to 100×100 m.

AEON Indonesia – retail

Jakarta, Indonesia

Management Trainee

Apr. 2018 – Sep. 2019

- Ran sales and operations analysis during a cross-functional rotation (store operations, merchandising, marketing, supply chain), turning store data into process-improvement recommendations.

SKK Migas – upstream oil & gas regulator

Jakarta, Indonesia

Data Analyst

Apr. 2016 – Apr. 2018

- Built an early ML model predicting oil & gas resources for 200+ working areas (<10% test error); migrated data from 50+ companies; built Python/Tableau dashboards.

SKILLS

Causal inference & econometrics: difference-in-differences, discrete choice (MNL, NL, ICLV) in Biogeme, factor analysis, welfare (logsum) estimation

Machine learning: XGBoost and gradient boosting, Random Forest, clustering, recommendation systems, credit scoring, CNNs, SHAP

Data & platforms: Python (pandas, polars, scikit-learn, statsmodels), R, SQL, PySpark / Spark ML, large-scale parquet pipelines, MLflow, Snowflake, Databricks, Dataiku, Airflow, AWS, GCP, Docker

Geospatial: QGIS, GeoPandas, OpenStreetMap, satellite imagery, POI and mobility data

LLMs: LLM prompt design and evaluation, LLM feature engineering (Llama-3), Snowflake Cortex, RAG (Elasticsearch), Text-to-SQL

LANGUAGES

English (advanced); Bahasa Indonesia (native)